

Ch. 1 — P01 (C) Classify Systems

Problem Bank | Chapter 1 | Type: C | Difficulty:

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For each system below, classify it along the three axes:

- **Discrete vs. Continuous** — does state change only at events, or continuously?
- **Stochastic vs. Deterministic** — does randomness play a role?
- **Static vs. Dynamic** — does the system evolve over time?

Then state which simulation paradigm (DES, continuous, agent-based, or none needed) you would use to study it.

Systems:

1. A checkout lane at a grocery store.
2. A water tank being filled and drained.
3. A car factory assembly line with fixed cycle times.
4. An emergency department with random patient arrivals.
5. The spread of a disease through a population where each individual makes random contact decisions.
6. A traffic light that switches on a fixed schedule.
7. A stock market order book.
8. A gas pipeline with pressure dynamics.

For each system, provide:

- The three-axis classification (2–3 words per axis).
- A brief justification (1 sentence each).
- Your recommended modeling paradigm.